$$\mathrm{e}^{-\hat{H}_{\mathcal{E}}^{(i)}/k_{\mathrm{B}}T}\hat{\sigma}_z^{(i)}(\tau)\hat{\sigma}_z^{(i)} = \mathrm{e}^{\Delta_i/2k_{\mathrm{B}}T - \mathrm{i}\Delta_i\tau}(|+\rangle\langle+|)_i + \mathrm{e}^{-\Delta_i/2k_{\mathrm{B}}T + \mathrm{i}\Delta_i\tau}(|-\rangle\langle-|)_i. \tag{5.182}$$